Continuous discharge (DP4.00130.001)

Measurement

Continuous discharge, the volumetric rate of water flowing through a stream at a given location, in liters per second. Data prior to RELEASE-2026 are reported as 1 minute instantaneous values. Beginning with RELEASE-2026, and all subsequent releases, data collected beginning with water year 2022 (Oct 1, 2021) are reported as 15-minute averages.

Collection methodology

Continuous discharge is derived from Elevation of surface water (DP1.20016.001), Discharge field collection (DP1.20048.001), Gauge height (DP1.20267.001), and Stage-discharge rating curves (DP4.00133.001). See documentation for those data products for sensor details and collection methodology.

For information about disturbances, land management activities, and other incidents that may impact data at NEON sites, see the Site management and event reporting (DP1.10111.001) data product.

Maintenance and calibration

Continuous discharge is derived from Elevation of surface water (DP1.20016.001), Discharge field collection (DP1.20048.001), Gauge height (DP1.20267.001), and Stage-discharge rating curves (DP4.00133.001). See documentation for those data products for sensor maintenance and calibration details.

Data package contents

csd_constantBiasShift: Information on the date range and magnitude of constant bias shifts applied to Instrumented Systems (IS) data streams during gap-filling
csd_continuousDischargeUSGS: Continuous discharge data from the nearby USGS station
csd_dataGapToFillMethodMapping: Summary of data gaps in cleaned Instrumented Systems (IS) data streams and the methods selected to gap-fill each data gap
csd_dischargeRegressionUSGS: Information relating NEON discharge readings to USGS discharge readings
csd_gapFillingRegression: Statistics and information on regressions developed for gap-filling Instrumented Systems (IS) data streams
csd_15_min: Continuous discharge data calculated from in situ pressure and stage-discharge rating curve averaged over 15 minutes
csd_gaugeWaterColumnRegression: Statistics and information on gauge height vs. water column height regressions used in estimating continuous discharge
sdrc_gaugePressureRelationship: Information relating staff gauge readings to pressure measurements
csd_continuousDischarge: Continuous discharge data calculated from in situ pressure and stage-discharge rating curve

variables: Description and units for each column of data in data tables
readme: Data product description, issue log, and other metadata about the data product
sensor_positions: Geospatial locations of individual sensors

Data processing and derivation

Continuous discharge is calculated using water column height data from the same calibrated pressure data used to produce Elevation of surface water (DP1.20016.001). An enhanced algorithm was implemented beginning with RELEASE-2025 and in all subsequent releases, applying to all Water Years starting with Water Year 2022 (from October 1, 2021). Water column height is cleaned by removing periods of erratic data. Shifts resulting from unintended movement of the pressure sensor are corrected. Gaps resulting from missing or cleaned data are then filled using the relationship with a secondary pressure sensor, or another parameter such as specific conductance. Indicator flags identifying which cleaning and gap-filling method was used are included in the expanded data package. Water column height is then converted to equivalent stage using a regression equation developed using the observed gauge heights (DP1.20048.001). Regression information is included in sdrc_gaugePressureRelationship and csd_gaugeWaterColumnRegression tables published in the expanded package of this data product.

The equivalent stage values are then converted to discharge by applying the Stage-discharge rating curve (DP4.00133.001) developed using a Bayesian model (Le Coz et al. 2014) which returns the maximum-likelihood (mode) of the posterior distribution. This maxpost discharge is then cleaned in a similar manner to water column height, with any remaining gaps potentially being filled using the relationship with external data sources such as collocated USGS or LTER sites. Indicator flags identifying which cleaning and gap-filling method was used are included in the expanded data package. The cleaned and gap-filled data is reported as continuous discharge. In releases prior to RELEASE-2025, for Water Years prior to Water Year 2022, and for provisional data that have not yet been reviewed, the process for calculating maxpost discharge from water column height is the same, but the inputs and outputs have not been cleaned or gap-filled.

Beginning with RELEASE-2026 and in all subsequent releases, water column height is averaged to a 15 minute resolution prior to correction, producing 15 minute averages of continuous discharge.

Data quality

Refer to the documentation for Elevation of surface water (DP1.20016.001), Discharge field collection (DP1.20048.001), Gauge height (DP1.20267.001), and Stage-discharge rating curves (DP4.00133.001) for information about data quality in the input data products. Continuous discharge data include both data flags and uncertainty estimates for the calculated stage and discharge. Beginning with RELEASE-2025 and in all subsequent releases, applying to all Water Years starting with Water Year 2022 (from October 1, 2021), when continuous discharge data cannot be cleaned and gap filled using the methods documented in the Data Processing section, values are set to NA. For data collected prior to Water Year 2022, and all water years in Releases prior to RELEASE-2025, maxpost discharge data identified as potentially suspect have the dischargeFinalQF set to 1. Estimated upper and lower bounds of parametric and remnant uncertainty from the Bayesian model are provided.

Please note that quality checks are comprehensive but not exhaustive; therefore, unknown data quality issues may exist. Also note that, conversely, some quality-flagged data are still usable depending on the scientific use case. Users are advised to evaluate quality of the data as relevant to the scientific research question being addressed, using the data quality flags, issue logs, and maintenance records included in download packages to aid interpretation.

Standard calculations

For wrapper functions to download data from the API, and functions to merge tabular data files across sites and months, NEON provides the neonUtilities package in R and the neonutilities package in Python. See the Download and Explore NEON Data tutorial for introductory instructions in both programming languages.

To merge 1 minute and 15 minute data tables, see the Continuous Discharge tutorial on the NEON code resources webpage.

Table joining

Table 1	Table 2	Join by field(s)
csd_continuousDischarge	csd_gaugeWaterColumnRegression	regressionID
csd_continuousDischargeUSGS	csd_dischargeRegressionUSGS	usgsDischargeRegID
csd_dataGapToFillMethodMapping	csd_gapFillingRegression	gapFillRegressionID
csd_continuousDischarge	sdrc_gaugePressureRelationship	Not fully automatable: Join by regressionID and endDate truncated to YYYY-MM-DD HH:MM (without seconds)
csd_continuousDischarge	csd_dataGapToFillMethodMapping	Not fully automatable: Join by csd_continuousDischarge: endDate overlapping with date range of records in csd_dataGapToFillMethodMapping
csd_continuousDischarge	csd_constantBiasShift	Not fully automatable: Join by csd_continuousDischarge: endDate overlapping with date range of records in csd_constantBiasShift

Table 1	Table 2	Join by field(s)
csd_continuousDischarge	csd_gapFillingRegression	Not fully automatable: Join by csd_continuousDischarge: endDate overlapping with date range of records in csd_gapFillingRegression
csd_dataGapToFillMethodMapping	csd_constantBiasShift	Join not recommended, see User Guide
csd_gapFillingRegression	csd_constantBiasShift	Join not recommended, see User Guide
csd_continuousDischarge	csd_continuousDischargeUSGS	Join not recommended, see User Guide
csd_continuousDischarge	csd_dischargeRegressionUSGS	Join not recommended, see User Guide
csd_continuousDischargeUSGS	sdrc_gaugePressureRelationship	Join not recommended, see User Guide

Documentation

AOS Protocol and Procedure: DSC – Stream Discharge
NEON.DOC.001085vJ | 2.4 MiB | PDF

NEON Aquatic Sampling Strategy
NEON.DOC.001152vD | 846.5 KiB | PDF

AOS Protocol and Procedure: GAG – Aquatic Staff Gauge Measurement Readings
NEON.DOC.005277vB | 1.1 MiB | PDF

AOS Standard Operating Procedure: DSC – Configurations, Settings, and Collection Methods for Stream Discharge Measurements Using the Flowmeter Method
NEON.DOC.005388vA | 643.8 KiB | PDF

AOS Standard Operating Procedure: DSC – Configurations, Settings, and Collection Methods for Stream Discharge Measurements using the ADCP Method
NEON.DOC.005389vA | 1.3 MiB | PDF

AIS Standard Operating Procedure: Conducting AIS Site Surveys at NEON Aquatic Sites
NEON.DOC.005399vA | 1.7 MiB | PDF

NEON Algorithm Theoretical Basis Document (ATBD): Stage-discharge rating curves and continuous discharge
NEON.DOC.005403vA | 551.2 KiB | PDF

NEON User Guide to Continuous Discharge (NEON.DP4.00130.001)
NEON_continuousQ_userGuide_vF | 257.4 KiB | PDF

For more information on data product documentation, see:
https://data.neonscience.org/data-products/DP4.00130.001

Citation

To cite data from value: "Continuous discharge" (DP4.00130.001), see citation here:
https://data.neonscience.org/data-products/DP4.00130.001
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